HE2001 Microeconomics II

Tutorial 1

Academic Year 2025/2026, Semester 2

Quantitative Research Society @NTU

August 17, 2026

Overview

This tutorial focuses on foundational two-good consumer theory and revealed preference reasoning:

- completeness, transitivity, and monotonicity ("increasingness") of a preference relation;
- budget lines, affordability, and geometric interpretation in $\mathbb{R}^2_+$;
- revealed preference comparisons across periods;
- why WARP violations cannot be rationalised by increasing preferences;
- constructing a (non-increasing / non-monotone) preference that can rationalise observed choices;
- limitations of the "chosen implies preferred" maxim in applications.

Question 1 (Lexicographic-type preference)

Problem

Consider the preference relation $\succeq$ on $\mathbb{R}^2$ defined by

$$(x_1, x_2) \succeq (y_1, y_2) \quad \Longleftrightarrow \quad \text{either (i) } x_1 > y_1, \text{ or (ii) } x_1 = y_1 \text{ and } x_2 \geq y_2.$$

Is this preference relation complete, transitive, and increasing?

Solution

Method 1: Direct verification from definitions

We recall the standard definitions.

- **Completeness:** for all x, y, either $x \succeq y$ or $y \succeq x$ (or both).
- **Transitivity:** for all x, y, z, if $x \succeq y$ and $y \succeq z$, then $x \succeq z$.
- **Increasing (monotone):** if $x \geq y$ componentwise and $x \neq y$, then $x \succ y$ (i.e. $x \succeq y$ and not $y \succeq x$).

Completeness. Fix any $x = (x_1, x_2)$ and $y = (y_1, y_2)$. Exactly one of the following holds:

$$x_1 > y_1, \qquad x_1 < y_1, \qquad x_1 = y_1.$$

If $x_1 > y_1$, then $x \succeq y$ by clause (i). If $x_1 < y_1$, then $y_1 > x_1$, hence $y \succeq x$ by clause (i). If $x_1 = y_1$, then either $x_2 \geq y_2$ or $y_2 \geq x_2$; in the first case $x \succeq y$ by clause (ii), and in the second case $y \succeq x$ by clause (ii). Thus $\succeq$ is complete.

Transitivity. Let $x, y, z \in \mathbb{R}^2$ satisfy $x \succeq y$ and $y \succeq z$. We prove $x \succeq z$ by cases.

- If $x_1 > y_1$, then necessarily $y_1 \geq z_1$ or $y_1 = z_1$; in either case $x_1 > z_1$, hence $x \succeq z$ by clause (i).
- If $x_1 = y_1$ and $x_2 \geq y_2$, then consider $y \succeq z$:
  - If $y_1 > z_1$, then $x_1 = y_1 > z_1$, so $x \succeq z$ by clause (i).
  - If $y_1 = z_1$ and $y_2 \geq z_2$, then $x_1 = y_1 = z_1$ and $x_2 \geq y_2 \geq z_2$, so $x \succeq z$ by clause (ii).

In all cases, $x \succeq z$. Hence $\succeq$ is transitive.

Increasingness. Let $x \geq y$ componentwise with $x \neq y$. Then either $x_1 > y_1$ or $x_1 = y_1$ and $x_2 > y_2$.

- If $x_1 > y_1$, then $x \succeq y$ by (i). Also $y \succeq x$ is impossible because it would require $y_1 > x_1$ or $y_1 = x_1$ with $y_2 \geq x_2$, contradicting $x_1 > y_1$. Hence $x \succ y$.
- If $x_1 = y_1$ and $x_2 > y_2$, then $x \succeq y$ by (ii). For $y \succeq x$ we would need $y_1 > x_1$ (false) or $y_1 = x_1$ and $y_2 \geq x_2$ (false since $x_2 > y_2$). Hence $x \succ y$.

Therefore $\succeq$ is increasing.

$$\boxed{\succeq \text{ is complete, transitive, and increasing.}}$$

Method 2: Identification with lexicographic order

Define the lexicographic order $\succeq_{\text{lex}}$ on $\mathbb{R}^2$ by

$$x \succeq_{\text{lex}} y \quad \iff \quad \big(x_1 > y_1\big) \text{ or } \big(x_1 = y_1 \text{ and } x_2 \geq y_2\big).$$

By inspection, the preference $\succeq$ is exactly $\succeq_{\text{lex}}$. We now verify the properties for $\succeq_{\text{lex}}$.

Completeness. For any x, y, if $x_1 \neq y_1$ then the larger first coordinate is lexicographically preferred; if $x_1 = y_1$, then the larger second coordinate is weakly preferred. Hence either $x \succeq_{\text{lex}} y$ or $y \succeq_{\text{lex}} x$.

Transitivity. Suppose $x \succeq_{\text{lex}} y$ and $y \succeq_{\text{lex}} z$. If $x_1 > y_1$, then $x_1 > z_1$ (since $y_1 \geq z_1$ or $y_1 = z_1$), so $x \succeq_{\text{lex}} z$. If $x_1 = y_1$ and $x_2 \geq y_2$, then either $y_1 > z_1$ (so $x_1 > z_1$) or $y_1 = z_1$ and $y_2 \geq z_2$ (so $x_2 \geq z_2$ with $x_1 = z_1$). In both cases $x \succeq_{\text{lex}} z$.

Increasingness. If $x \geq y$ and $x \neq y$, then either $x_1 > y_1$ or $x_1 = y_1$ and $x_2 > y_2$; either way $x \succ_{\text{lex}} y$. Hence increasingness holds.

$$\boxed{\succeq \text{ is lexicographic-type: complete, transitive, and increasing.}}$$

Question 2 (Budget lines and affordability)

Problem

Draw the budget line for prices $p^{(1)} = (1, 2)$ and $p^{(2)} = (2, 1)$ when expenditure is $m = 2$. Draw the consumption bundle $x = (1, 1)$. Is this bundle affordable at either budget?

Solution

Method 1: Geometric check (point lies above each line)

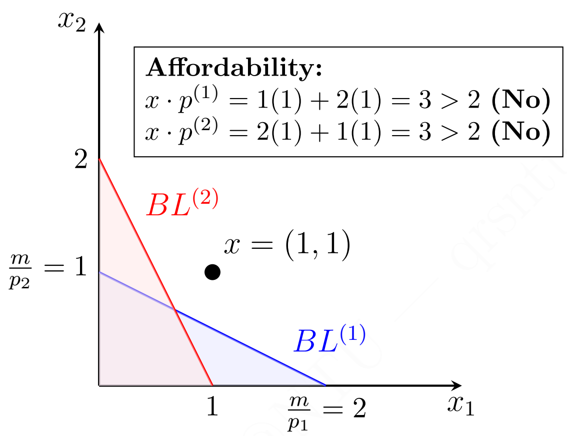

Figure 1: Budget sets for $p^{(1)} = (1, 2)$ and $p^{(2)} = (2, 1)$ with $m = 2$. The bundle $x = (1, 1)$ lies outside both budget sets.

For $p^{(1)}$, feasibility on/under the line requires $x_1 + 2x_2 \leq 2$. At $(1, 1)$,

$$1 + 2(1) = 3 > 2,$$

so $(1, 1)$ lies above that line.

For $p^{(2)}$, feasibility on/under the line requires $2x_1 + x_2 \leq 2$. At $(1, 1)$,

$$2(1) + 1 = 3 > 2,$$

so $(1, 1)$ lies above that line too.

$$\boxed{\text{Graphically, } x = (1, 1) \text{ lies above both budget lines, hence is infeasible.}}$$

Method 2: Write the budget line equations and evaluate costs

For a two-good economy, the budget line with prices $p = (p_1, p_2)$ and expenditure m is

$$p_1 x_1 + p_2 x_2 = m.$$

Budget line for $p^{(1)} = (1, 2)$, $m = 2$. The equation is

$$x_1 + 2x_2 = 2.$$

Intercepts: $(2, 0)$ and $(0, 1)$.

Budget line for $p^{(2)} = (2, 1)$, $m = 2$. The equation is

$$2x_1 + x_2 = 2.$$

Intercepts: $(1, 0)$ and $(0, 2)$.

Affordability of $x = (1, 1)$. Compute expenditures:

$$p^{(1)} \cdot x = 1 \cdot 1 + 2 \cdot 1 = 3, \qquad p^{(2)} \cdot x = 2 \cdot 1 + 1 \cdot 1 = 3.$$

Since $3 > 2$, the bundle is not affordable under either budget.

$$\boxed{x = (1, 1) \text{ is not affordable under either budget.}}$$

Question 3 (Revealed preference across periods)

Problem

Suppose a preference-maximising consumer in period 1 buys $x^1 = (2,2)$ when prices are $p^1 = (1,1)$, and buys $x^2 = (0,3)$ in period 2 when prices are $p^2 = (2,1)$. In which period is the consumer better off (i.e. in which period does the consumer obtain a better bundle)?

Solution

Method 1: Direct revealed-preference comparison via affordability

Let $m^t := p^t \cdot x^t$ be the expenditure in period t. Compute:

$$m^1 = p^1 \cdot x^1 = 1 \cdot 2 + 1 \cdot 2 = 4, \qquad m^2 = p^2 \cdot x^2 = 2 \cdot 0 + 1 \cdot 3 = 3.$$

Check whether x^2 was affordable in period 1:

$$p^1 \cdot x^2 = 1 \cdot 0 + 1 \cdot 3 = 3 \le 4,$$

so x^2 was feasible when x^1 was chosen. Therefore, period-1 optimality implies

$$x^1 \succeq x^2.$$

Check the reverse affordability in period 2:

$$p^2 \cdot x^1 = 2 \cdot 2 + 1 \cdot 2 = 6 > 3,$$

so x^1 was not feasible in period 2; hence period 2 cannot reveal $x^2 \succeq x^1$.

$$\boxed{\text{The consumer is (weakly) better off in period 1: } x^1 \succeq x^2.}$$

Method 2: Cross-expenditure inequalities ("price-index" view)

From period 1, choosing x^1 at p^1 means every bundle with weakly lower period-1 cost cannot be strictly preferred to x^1:

$$p^1 \cdot x \le p^1 \cdot x^1 \implies x^1 \succeq x.$$

Since

$$p^1 \cdot x^2 = 3 \le 4 = p^1 \cdot x^1,$$

we conclude $x^1 \succeq x^2$.

Meanwhile,

$$p^2 \cdot x^1 = 6 > 3 = p^2 \cdot x^2,$$

so period 2 provides no basis to infer $x^2 \succeq x^1$.

$$\boxed{x^1 \succeq x^2 \text{ is the only welfare conclusion supported by the data.}}$$

Question 4 (WARP violation and impossibility under increasing preferences)

Problem

Consider a consumer who buys $x^1 = (1, 2)$ when prices are $p^1 = (1, 2)$ and buys $x^2 = (2, 1)$ when prices are $p^2 = (2, 1)$. The consumer does not satisfy WARP. Prove that there is no increasing preference $\succeq$ which can explain the consumer's behaviour. That is, show that there is no increasing $\succeq$ such that

$$x^1 \succeq x \text{ for all } x \in B(p^1,\, p^1 \cdot x^1), \qquad x^2 \succeq x \text{ for all } x \in B(p^2,\, p^2 \cdot x^2).$$

Solution

Method 1: Contradiction via strict improvement from "slack"

Compute the expenditures:

$$m^1 = p^1 \cdot x^1 = 1 \cdot 1 + 2 \cdot 2 = 5, \qquad m^2 = p^2 \cdot x^2 = 2 \cdot 2 + 1 \cdot 1 = 5.$$

Step 1: Mutual affordability forces mutual weak preference. Cross-affordability:

$$p^1 \cdot x^2 = 1 \cdot 2 + 2 \cdot 1 = 4 \le 5 \quad \Rightarrow \quad x^1 \succeq x^2,$$

$$p^2 \cdot x^1 = 2 \cdot 1 + 1 \cdot 2 = 4 \le 5 \quad \Rightarrow \quad x^2 \succeq x^1.$$

So any rationalising $\succeq$ would imply $x^1 \sim x^2$.

Step 2: Increasingness yields a feasible bundle strictly better than x^2 in period 1. Because $p^1 \cdot x^2 = 4 < 5$, the bundle x^2 leaves strictly positive expenditure unused under p^1. Use the remaining 1 unit of expenditure to increase good 1 (priced at 1 under p^1):

$$y := x^2 + (1, 0) = (3, 1).$$

Then

$$p^1 \cdot y = p^1 \cdot x^2 + p^1 \cdot (1, 0) = 4 + 1 = 5,$$

so $y \in B(p^1, m^1)$. Also $y \ge x^2$ and $y \ne x^2$, hence by increasingness $y \succ x^2$.

Step 3: Period-1 optimality forces $x^1 \succ x^2$, contradicting period-2 optimality. Since $y \in B(p^1, m^1)$ and x^1 is optimal there, $x^1 \succeq y$. With $y \succ x^2$ and transitivity, $x^1 \succ x^2$. But $x^1 \succ x^2$ means *not* $x^2 \succeq x^1$, contradicting the earlier implication $x^2 \succeq x^1$.

$$\boxed{\text{No increasing preference } \succeq \text{ can rationalise these two choices.}}$$

Method 2: General lemma — increasing preferences imply WARP

Lemma. Assume $\succeq$ is increasing and choices x^1, x^2 are $\succeq$-maximisers on their respective budget sets. Then it is impossible to have both $p^1 \cdot x^2 \leq p^1 \cdot x^1$ and $p^2 \cdot x^1 \leq p^2 \cdot x^2$ with at least one inequality strict and $x^1 \neq x^2$. (That is, increasing rationalisable data must satisfy WARP.)

Proof. Assume for contradiction that x^2 is affordable when x^1 is chosen and x^1 is affordable when x^2 is chosen, with (say) $p^1 \cdot x^2 < p^1 \cdot x^1$. Then optimality gives $x^1 \succeq x^2$ and $x^2 \succeq x^1$, so $x^1 \sim x^2$.

Let $\Delta := p^1 \cdot x^1 - p^1 \cdot x^2 > 0$. Because prices are positive, pick $i \in \{1, 2\}$ and $\varepsilon > 0$ such that $p_i^1 \varepsilon \leq \Delta$. Set $y := x^2 + \varepsilon e_i$. Then $y \geq x^2$, $y \neq x^2$, and

$$p^1 \cdot y = p^1 \cdot x^2 + p_i^1 \varepsilon \leq p^1 \cdot x^2 + \Delta = p^1 \cdot x^1,$$

so y is feasible in period 1. Increasingness gives $y \succ x^2$, and period-1 optimality gives $x^1 \succeq y$, hence $x^1 \succ x^2$, contradicting $x^2 \succeq x^1$.

Apply to the data. Here $p^1 \cdot x^2 = 4 < 5 = p^1 \cdot x^1$ and $p^2 \cdot x^1 = 4 < 5 = p^2 \cdot x^2$, so the lemma gives a contradiction immediately.

$\boxed{\text{Since the choices violate WARP, no increasing rationalising preference exists.}}$

Question 5 (Constructing a non-increasing preference that rationalises the choices)

Problem

Define a non-increasing preference $\succeq$ which can explain the behaviour of the consumer in the previous question.

Solution

Method 1: Norm-minimising ("smaller bundles are better") preference

Define $\succeq$ on $\mathbb{R}^2_+$ by

$$x \succeq y \quad \iff \quad \|x\|_2 \leq \|y\|_2,$$

equivalently via the utility $u(x) := -\|x\|_2^2 = -(x_1^2 + x_2^2)$.

Non-increasingness. If $x \geq y$ componentwise, then $x_1^2 \geq y_1^2$ and $x_2^2 \geq y_2^2$, hence

$$\|x\|_2^2 \geq \|y\|_2^2 \quad \Rightarrow \quad y \succeq x.$$

So "more of both goods" cannot improve welfare; the preference is non-increasing.

Rationalisation on the budget lines of Question 4. In Question 4, $m^1 = m^2 = 5$ and the budget lines are

$$x_1 + 2x_2 = 5, \qquad 2x_1 + x_2 = 5.$$

Under $u(x) = -\|x\|_2^2$, the consumer chooses the bundle on each line that *minimises* $\|x\|_2^2$.

Period 1. Minimise $x_1^2+x_2^2$ subject to $x_1+2x_2 = 5$. Lagrangian $\mathcal{L} = x_1^2+x_2^2+\lambda(x_1+2x_2-5)$. FOCs:

$$2x_1 + \lambda = 0, \qquad 2x_2 + 2\lambda = 0.$$

So $\lambda = -2x_1$ and $2x_2 - 4x_1 = 0 \Rightarrow x_2 = 2x_1$. Impose the constraint: $x_1 + 2(2x_1) = 5 \Rightarrow 5x_1 = 5 \Rightarrow x_1 = 1$, hence $x_2 = 2$. Thus $x^1 = (1, 2)$ is optimal on the first budget line.

Period 2. Minimise $x_1^2+x_2^2$ subject to $2x_1+x_2 = 5$. Lagrangian $\mathcal{L} = x_1^2+x_2^2+\lambda(2x_1+x_2-5)$. FOCs:

$$2x_1 + 2\lambda = 0, \qquad 2x_2 + \lambda = 0.$$

Thus $\lambda = -x_1 = -2x_2 \Rightarrow x_1 = 2x_2$. Impose the constraint: $2(2x_2) + x_2 = 5 \Rightarrow 5x_2 = 5 \Rightarrow x_2 = 1$, hence $x_1 = 2$. Thus $x^2 = (2, 1)$ is optimal on the second budget line.

$$\boxed{\text{One rationalising non-increasing preference is } u(x) = -(x_1^2 + x_2^2).}$$

Method 2: Cauchy–Schwarz (closed-form optimiser on a budget line)

For any $p \in \mathbb{R}^2_{++}$ and $m > 0$, consider the budget line $p \cdot x = m$. By Cauchy–Schwarz,

$$(p \cdot x)^2 \leq \|p\|_2^2 \, \|x\|_2^2.$$

If $p \cdot x = m$, then

$$\|x\|_2^2 \geq \frac{m^2}{\|p\|_2^2},$$

with equality if and only if x is a nonnegative scalar multiple of p, i.e. $x = \alpha p$. Imposing $p \cdot x = m$ yields $\alpha = m/\|p\|_2^2$, hence the unique norm-minimiser is

$$x^\star = \frac{m}{\|p\|_2^2} \, p.$$

Apply with $m = 5$.

Period 1. $p^1 = (1, 2)$, $\|p^1\|_2^2 = 1^2 + 2^2 = 5$, so

$$x^\star = \frac{5}{5}(1, 2) = (1, 2) = x^1.$$

Period 2. $p^2 = (2, 1)$, $\|p^2\|_2^2 = 2^2 + 1^2 = 5$, so

$$x^\star = \frac{5}{5}(2, 1) = (2, 1) = x^2.$$

$$\boxed{x^t = \frac{m^t}{\|p^t\|_2^2} \, p^t \text{ under } u(x) = -\|x\|_2^2, \quad t = 1, 2.}$$

Question 6 (When "chosen implies preferred" can be problematic)

Problem

The first maxim of classic consumer theory states that *"What is chosen is preferred to what could have been chosen."* Can you think of some situations where this maxim may be problematic?

Solution

Method 1: The analyst's "could have been chosen" set may be wrong

The maxim assumes the analyst correctly observes the decision maker's true feasible set. In practice, a budget constraint used by the analyst may omit real constraints faced by the consumer.

Examples:

- **Availability constraints:** stockouts, rationing, or per-customer limits mean some "affordable" bundles were not actually feasible.
- **Transaction/search costs:** time, transportation, fees, or information frictions reduce the effective opportunity set.
- **Indivisibilities:** if goods come in integer units, many points on a continuous budget line cannot be chosen.
- **Institutional constraints:** eligibility rules, contracts, penalties, or required bundles restrict feasible options.

Method 2: Observed choice may not reveal stable preference maximisation

Even if feasibility is correctly measured, observed choice can be contaminated by factors outside a stable preference-maximisation model.

Examples:

- **Mistakes/limited attention:** the consumer may choose a dominated option due to errors or cognitive constraints.
- **Uncertainty/incomplete information:** ex ante choice may not align with ex post preference comparisons.
- **Time inconsistency:** short-run and long-run selves may disagree, so choice may not reflect the welfare-relevant preference.
- **Social/strategic motives:** signalling, norms, or incentives can drive choice in ways not captured by the model.